Barry V. Martin

LEAD DATA ENGINEER | AIOPS, MLOPS & GENERATIVE AI

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PROFESSIONAL PROFILE

Lead Data Engineer specializing in **AIOps, MLOps, and production-grade generative AI**. Builds reliable AI operations through evaluation, model monitoring, automated alerts, and governed inference workflows across AWS, Snowflake, and Apache Airflow. Partners with data scientists, engineers, and business stakeholders to turn AI and machine learning use cases into secure, maintainable production systems.

PROFESSIONAL EXPERIENCE

Lead Data Engineer (MLOps) | Zelis

Remote | Aug 2021 - Present

SELECTED AI/ML PLATFORMS

- **GenAI Evaluation Platform:** Developed an enterprise platform for evaluating company GenAI applications and custom GPTs, including a reusable Python metrics library, AWS Step Functions ETL workflows, and an AWS-hosted Streamlit dashboard with SSO and role-based access control.
- **Model Monitoring Platform:** Designed daily data drift and model performance monitoring with Airflow, AWS Lambda, Amazon ECS, a custom Snowflake data model, Streamlit reporting, and automated email alerts for critical model issues.
- **Batch Inference Pipelines:** Built Airflow-orchestrated SageMaker and SageMaker Pipelines workflows for traditional ML inference, persisting prediction outputs to Snowflake for reporting and downstream analysis.
- **Agentic Jira Assistant:** Created a Streamlit application powered by a custom AI agent that uses project context and brief user requests to generate structured Jira descriptions, acceptance criteria, and implementation details.
- **ChatGPT Logging & Secure Explorer:** Implemented daily Amazon ECS jobs using the OpenAI Compliance API to retrieve Enterprise conversations, store standardized JSON in Amazon S3, generate custom GPT datasets, and provide authorized users an authenticated search, view, and download interface.

PLATFORM ENGINEERING, GOVERNANCE & DELIVERY

- **Security & Access Governance:** Implemented SSO, role-based access control, and restrictive AWS permissions to protect sensitive enterprise AI data and internal applications.
- **Cloud Infrastructure & Delivery:** Automated repeatable application and platform deployments with Terraform, Docker, Amazon ECR, ECS, Lambda, IAM, SSO, and CI/CD workflows.
- **Large-Scale Data Processing:** Developed Spark and Amazon EMR pipelines to process high-volume datasets and support scalable analytics and machine learning workflows.
- **Snowflake & Airflow Engineering:** Designed database structures and production data pipelines that integrate Snowflake with Apache Airflow for dependable, repeatable processing.
- **Cross-Functional Delivery:** Partnered with data scientists, engineers, and business stakeholders to productionize models and deliver secure, maintainable AI/ML systems.

Senior Data Analyst | C.H. Robinson

Remote | Jan 2014 - Aug 2021

- Conducted exploratory analysis in Jupyter to identify trends and deliver actionable business insights.
- Built interactive analytics applications and dashboards with Plotly, Streamlit, Power BI, and R Shiny.
- Optimized SQL queries and data workflows across SQL Server, PostgreSQL, and Hive environments.
- Developed Python-based ETL, data integration, and automated reporting solutions.
- Improved data quality by integrating and validating disparate data sources and translating stakeholder requirements into scalable solutions.

TECHNICAL SKILLS

PROGRAMMING & DATA ENGINEERING

Python, SQL, Apache Airflow, Spark, Docker, Bash, ETL/ELT, API integration

GENERATIVE AI, AIOPS & MLOPS

GenAI evaluation, AI agents, model monitoring, data drift, automated alerts, batch inference, usage analytics

AWS

SageMaker, Step Functions, ECS, ECR, Lambda, S3, IAM, RDS, Glue, EMR

DATA PLATFORMS

Snowflake, PostgreSQL, SQL Server, Hive, database design, data modeling

APPLICATIONS & VISUALIZATION

Streamlit, Flask, Jupyter, Plotly, Power BI, R Shiny, Mermaid

INFRASTRUCTURE & CI/CD

Terraform, GitHub Actions, GitHub Enterprise, SSO, role-based access control

AI-AUGMENTED ENGINEERING PRACTICES

- **AI Pair Programming:** Use Cursor, GitHub Copilot, and Codex to accelerate test generation, boilerplate, refactoring support, and routine engineering tasks while maintaining review ownership of production code.
- **Agent-Assisted Documentation:** Direct coding agents to draft technical documentation, summarize implementation details, and explain completed work to improve maintainability, handoffs, and team knowledge sharing.

EDUCATION

University of Massachusetts Lowell

B.S. Information Technology

Focus: programming and relational databases

University of Oklahoma

Programming and relational database coursework

Research assistant, NSSL

Minor in Mathematics